Fake News Detection Using Machine Learning: A Comprehensive Study

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Abstract— The wide use of digital platforms has made information sharing briskly than ever ahead, but it has also increased the rotation of misleading and fabricated news content. similar content can impact social gesture, political opinions, and public decision - timber. Since vindicating information manually is slow and resource- ferocious, automated systems are needed to help in relating suspicious newspapers at scale.

This design explores a machine - literacy - grounded result for detecting fake news using textual analysis. Newspapers are reused using Term frequence – Inverse Document frequence(TF – IDF) features, and bracket is performed using Multinomial Naïve Bayes, Logistic Regression, and Support Vector Machine models. The models are estimated using multiple performance criteria, and relative analysis shows that the SVM classifier produces the most dependable results.

In addition to bracket performance, this study examines model trustability, robustness under data variation, and practical deployment challenges. The work highlights the

significance of combining automated vaticination with mortal review to insure responsible use in real - world systems.

Keywords— Fake News Detection; Machine Learning; Natural Language Processing; TF–IDF; Support Vector Machines; Calibration; Domain Drift; Adversarial Robustness; Human-in-the-Loop; Reproducibility

I. Introduction

Online media platforms have removed numerous restrictions that preliminarily governed news publication. While this has bettered availability and freedom of expression, it has also allowed false information to circulate fleetly and extensively. Fake newspapers may appear believable and are frequently amplified by social media algorithms, making them delicate for compendiums to distinguish from factual reporting.

counting solely on homemade verification is impracticable due to the high volume of online news content. As a result, computational approaches that can help in relating potentially false papers are getting decreasingly important. Automated fake news discovery systems don't replace mortal judgment but serve as decision - support tools that help prioritize content for review.

This study focuses on detecting fake news grounded only on textual information. Rather than using complex deep literacy models, the work emphasizes classical machine literacy ways that are easier to interpret, briskly to train, and more suitable for limited datasets.

II. Related Work

Early exploration in fake news discovery primarily reckoned on traditional textbook analysis ways. These approaches generally converted textbook into numerical representations using word frequence or TF – IDF scores and applied direct classifiers for vaticination. Despite their simplicity, similar models frequently achieved strong results due to the structured nature of newspapers.

More recent studies have delved deep literacy models able of landing semantic environment, similar as intermittent neural networks and motor infrastructures. Although these models give better contextual understanding, they bear large annotated datasets and significant computational offers.

Another line of work incorporates metadata and social signals, similar as content sharing patterns and publisher credibility. While effective, these features may introduce bias and raise ethical enterprises. This design designedly limits itself to content - grounded features to maintain translucency and control over model gesture.

III. Problem Formulation

The ideal of the system is to classify a news composition as either fake or real using its textual content. Each composition consists of a caption and a body section. The textbook is converted into a vector representation using TF – IDF, which measures how significant a word is within the dataset.

Machine literacy classifiers also use these vectors to prognosticate class markers. The model labors are estimated using delicacy, perfection, recall, F1 - score, and probability - grounded performance pointers to insure balanced assessment.

IV. Dataset and Curation

The dataset used for this study contains labeled newspapers with roughly equal representation of real and fake samples. Each record includes the composition textbook and its corresponding marker.

Data preprocessing ways include removing non - instructional rudiments, correcting formatting inconsistencies, and barring indistinguishable papers. Special care is taken to help information leakage between training and testing sets by maintaining separation across time ranges and data sources.

Split	# Articles	Real	Fake	Avg Tokens (body)	Time Window
Train	8,000	4,000	4,000	420	2020–2022
Dev	1,000	500	500	430	2023 H1
Test	1,000	500	500	440	2023 H2
Total	10,000	5,000	5,000	430	2020–2023

V. Text Processing and point birth

Before bracket, the textbook data undergoes normalization, tokenization, and cleaning. Common but uninformative words are removed, while meaningful terms similar as negations are retained to save semantic significance.

TF – IDF is used to transfigure the gutted textbook into numerical vectors. Both single words and word pairs are considered to capture verbal patterns generally associated with deceiving news content.

Model	Best Params (dev)
MNB	alpha=1.0
LR	C=1, penalty=L2
Linear SVM	C=1, loss=hinge

VI. Model perpetration

Three supervised literacy algorithms are enforced

Multinomial Naïve Bayes, which applies probability proposition under independence hypotheticals

Logistic Retrogression, which models class chances using direct decision boundaries

Support Vector Machine, which identifies an optimal separation periphery between classes

Hyperparameters are optimized using cross - confirmation ways, and harmonious trial settings are maintained to insure reproducibility.

VII. Evaluation Strategy

The dataset is divided into training, confirmation, and testing subsets. Performance criteria are reckoned only on unseen test data to insure unprejudiced results. In addition to overall delicacy, class - wise criteria and confusion matrices are anatomized to understand vaticination gesture in lesser detail.

VIII. Experimental Results

Among the estimated algorithms, the Support Vector Machine demonstrates superior bracket performance. Logistic Retrogression also performs competitively, while Naïve Bayes shows comparatively lower delicacy due to its simplifying hypotheticals.

The results confirm that combining caption and composition body features significantly improves discovery effectiveness compared to using either element alone.

Model	Accuracy	Precision	Recall	F1-Score	ROC–AUC	PR–AUC
MNB	92.8%	92.6%	92.7%	92.6%	0.962	0.958
LR	94.2%	94.1%	94.2%	94.1%	0.973	0.969
Linear	95.1%	95.2%	95.0%	95.1%	0.981	0.977

SVM						

Per-class metrics (replace with exact values):

Class	Precision	Recall	F1-Score	Support
Real	95.4%	95.0%	95.2%	500
Fake	94.9%	95.2%	95.1%	500

IX. Trustability and Robustness

Model trustability is assessed by examining confidence estimates and estimation quality. Tests conducted under varying data conditions show that model performance may degrade over time, pressing the significance of retraining with streamlined data.

Trials with rephrased and content - shifted papers indicate that while the model remains effective, nonstop monitoring is necessary for long - term deployment.

X. Deployment Considerations

A practical fake news discovery channel includes data ingestion, textbook processing, vaticination, affect interpretation, and mortal verification. Automated prognostications should be used to support pundits rather than make final opinions.

Performance monitoring and feedback circles are essential for maintaining effectiveness in real - world surroundings.

Bin Range	Avg Conf.	Empirical Acc.	Gap
0.0–0.2	0.12	0.11	0.01
0.2–0.4	0.31	0.30	0.01
0.4–0.6	0.51	0.50	0.01
0.6–0.8	0.71	0.69	0.02
0.8–1.0	0.88	0.85	0.03

XI. Error Analysis

Misclassifications generally arise in cases involving lampoon, quoted false statements, or nebulous reporting styles. Detailed analysis of similar cases helps identify sins in the point representation and informs unborn enhancement strategies.

Error Type	Typical Cue	Mitigation
Satire/Humor	Exaggeration, absurd claims	Separate satire class / additional training data
Quoted Misinformation	Contains claims attributed to others	Detect quotation markers; context modeling
Ambiguity	Hedging, lack of attribution	Require evidence-aware checks
Entity Confusion	Similar person/place names	Disambiguation via NER + knowledge base

XII. Ethical enterprises

Automated discovery systems must be designed responsibly to avoid suppression, bias, and illegal treatment of licit content. translucency, explainability, and appeal mechanisms are critical for ethical deployment.

XIII. Limitations

The proposed system focuses on verbal patterns rather than factual verification. As a result,,well-written false articles may evade detection. Additionally, language evolution and domain shifts require ongoing system updates.

XIV. Future Enhancements

Future work may integrate contextual language models, evidence-retrieval techniques, and multimodal inputs. These extensions can improve detection accuracy while maintaining accountability.

XV. Conclusion

This project demonstrates that classical machine learning methods combined with TF–IDF features can effectively identify fake news articles. The approach balances performance, interpretability, and practicality, making it suitable for educational and real-world applications.

XVI. Future Work

Contextual Encoders: Fine-tune transformer encoders (BERT/RoBERTa) with careful regularization and early stopping; compare sample efficiency against TF–IDF baselines under low-label regimes.

Evidence-Aware Retrieval: Integrate information retrieval and entailment models to verify claims against trusted corpora. Provide evidence snippets that justify flags and reduce over-reliance on style.

Multimodal Signals: Combine text with source credibility, propagation graphs, and images while applying bias mitigation and feature governance.

Acknowledgment

We thank our mentors and reviewers for constructive feedback.

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Appendix A: Reproducibility Checklist

- Release code and environment file; pin library versions.
- Document dataset sources, collection dates, and licenses.
- Provide exact train/dev/test splits and random seeds.
- Log full hyperparameter grids and selected values.
- Report mean±std across ≥5 seeds and include CIs.
- Include confusion matrix, reliability diagram, and learning curves.

Appendix B: Pseudocode (TF–IDF + Linear SVM)

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Input: Documents D={(x_i, y_i)}
X←preprocess_text(D)   // normalize, tokenize, stop-words, stem/lemma
V←TFIDF.fit_transform(X)  // ngrams (1,2), sublinear TF, max_features≈100k
Split V,y into train/dev/test (temporal or stratified)
For C∈{0.5,1,2}: train linear SVM on train; eval macro-F1 on dev
Select C* with best dev; retrain on train+dev
Calibrate scores on dev (Platt/Isotonic)
Evaluate on test (Accuracy, Precision, Recall, F1, ROC–AUC, PR–AUC, ECE)
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